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Mortality and Inequalities Research Group

LONDON SCHOOL of HYGIENE &TROPICAL MEDICINE

Kinship demography II

How to quantify orphanhood inequality in the US

Yu Chen & José Manuel Aburto

06/03/2026

Chen's previous work

THE LANCET

Global minimum estimates of children affected by COVID-19-associated orphanhood and deaths of caregivers: a modelling study

Susan D Hillis*, H Juliette T Unwin*, Yu Chen*, Lucie Cluver, Lorraine Sherr, Philip S Goldman, Oliver Rat
Andrés Villaveces, Alexander Butchart, Gretchen Bachman, Laura Rawlings, Phil Green, Charles A Nelsor

<== During the pandemic

Global estimation paper

Longitudinal analysis ==>

US orphanhood paper

nature medicine

Article

https://doi.org/10.1038/s41591-024-03343-6

Orphanhood and caregiver death among children in the United States by all-cause mortality, 2000–2021

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Check for updates

Andrés Villaveces[1,11], Yu Chen[2,11], Sydney Tucker[2,3], Alexandra Blenkinsop[2], Lucie Cluver[3,4], Lorraine Sherr[5], Jan L. Losby[1], Linden Graves[2], Rita Noonan[1], Francis Annor[1], Victor Kojey-Merle[2], Douhan Wang[2], Greta Massetti[1], Laura Rawlings[6], Charles A. Nelson[7], H. Juliette T. Unwin[8], Seth Flaxman[9], Susan Hillis[2,3,10,11] & Oliver Ratmann[2,11]

Discussion

Orphanhood and caregiver death among children in the United States by all-cause mortality, 2000–2021
https://doi.org/10.1038/s41591-024-03343-6

Questions:

Main takeaway messages?

Data sources and limitations?

The top five causes of death drive orphanhood incidence? Are they the same across race and ethnicity? Same across US states?

Which part of the paper do you think can be improved?

Motivation

An advanced topic – inequality analysis

Persistent **racial and ethnic inequalities** in **orphanhood** **timing** in the US

- Why **orphanhood**: Parental mortality is a major shock to children.
- Why **racial/ethnic inequalities**: SES disadv. and disparities.
- Why **orphanhood timing**: Losing a parent early in life differ that during adolescence.

Others' work studying U.S. data

- Orphanhood due to drug poisoning and firearm violence[1].
- Bereavement multiplier of kins due to COVID-19[2].
- …

[1] Schluter, B. S., Alburez-Gutierrez, D., Bibbins-Domingo, K., Alexander, M. J. & Kiang, M. V. Youth experiencing parental death due to drug poisoning and firearm violence in the US, 1999–2020. JAMA 331, 1741–1747 (2024).

[2] Verdery, A.M., Smith-Greenaway, E., Margolis, R. and Daw, J., 2020. Tracking the reach of COVID-19 kin loss with a bereavement multiplier applied to the United States. *Proceedings of the National Academy of Sciences*, *117*(30), pp.17695-17701.

Interpretation and contribution

Mortality regimes can reflect orphanhood inequality

- ***Cause of death***:

  is associated with *behaviour and environment*;

  can translate into *unequal risk* to orphanhood;

  can alter orphanhood patterns across *race and ethnicity* in the US.

Objective

- Time trends in orphanhood incidence by race and ethnicity.
- Behaviour related causes of death drive orphanhood incidence.
- Cause of death reshapes disparity patterns of orphanhood timing.

Method

- Data: vital statistics in the US [1]

- Statistical framework:
  - Orphanhood estimation is processed based on previous pipeline [2].
  - Disparity measurement: non-overlap metric (S index) [3].
  - Decomposition method: Shapley value function [4].

[1] National Center for Health Statistics. Data Access - Vital Statistics Online (2023). Accessed Aug 3, 2023. https://www.cdc.gov/nchs/data_access/vitalstatsonline.htm.
[2] Villaveces, A., Chen, Y. et al. Orphanhood and caregiver death among children in the United States by all-cause mortality, 2000–2021. Nature Medicine 31, 672–683 (2025) .
[3] Shi, J., Aburto, J. M., Martikainen, P., Tarkiainen, L. & van Raalte, A. A distributional approach to measuring lifespan stratification. Population Studies 77, 15–33 (2023)
[4] Shapley, L. S. et al. A value for n-person games (1953)

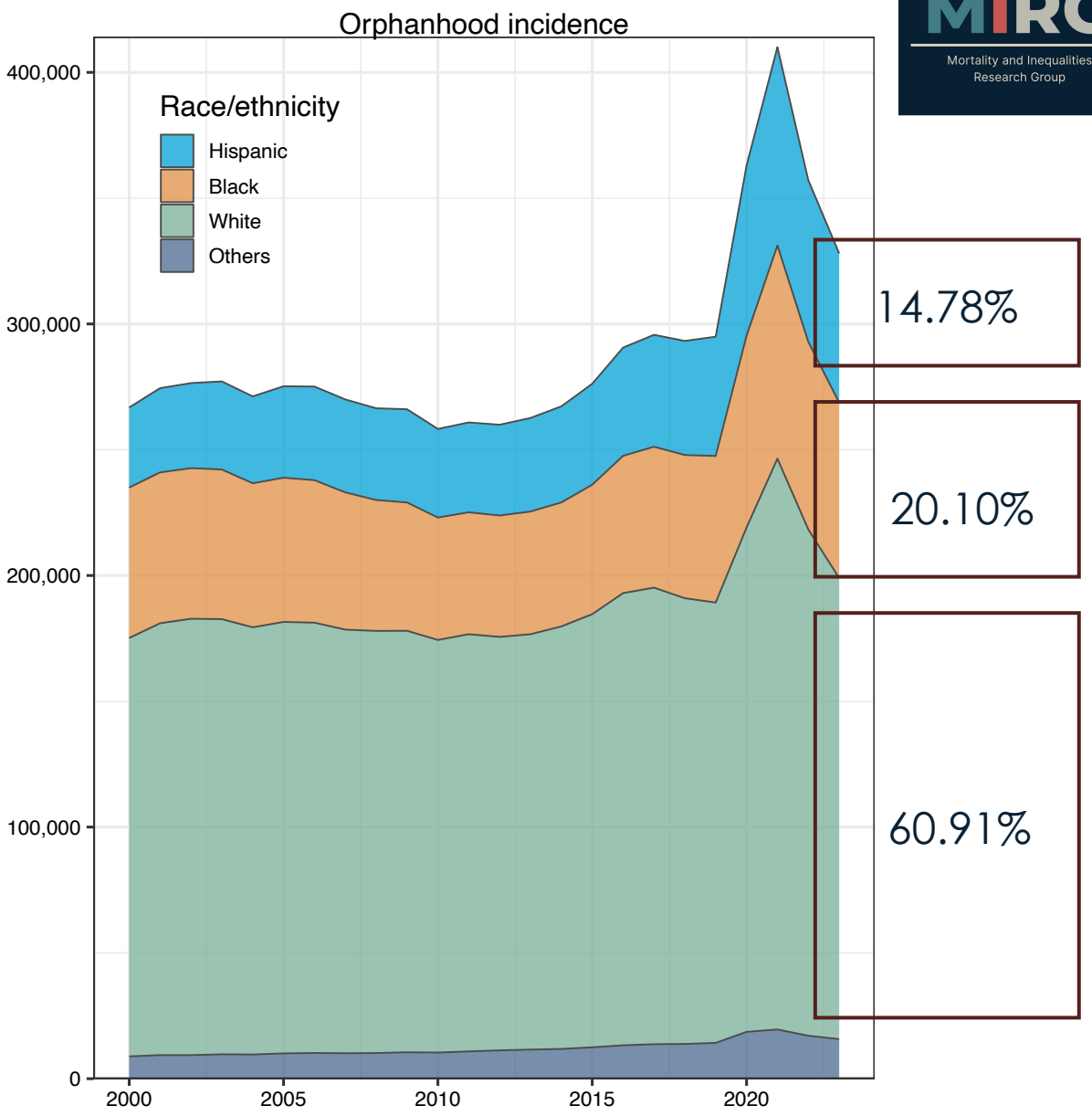
Assumptions

- Identical Race and ethnicity assumption in the same household (parents and children)
  - Hispanic
  - Non-Hispanic Black (Black in short)
  - Non-Hispanic White (White in short)
  - Others

- Homogeneous assumption on fertility rates

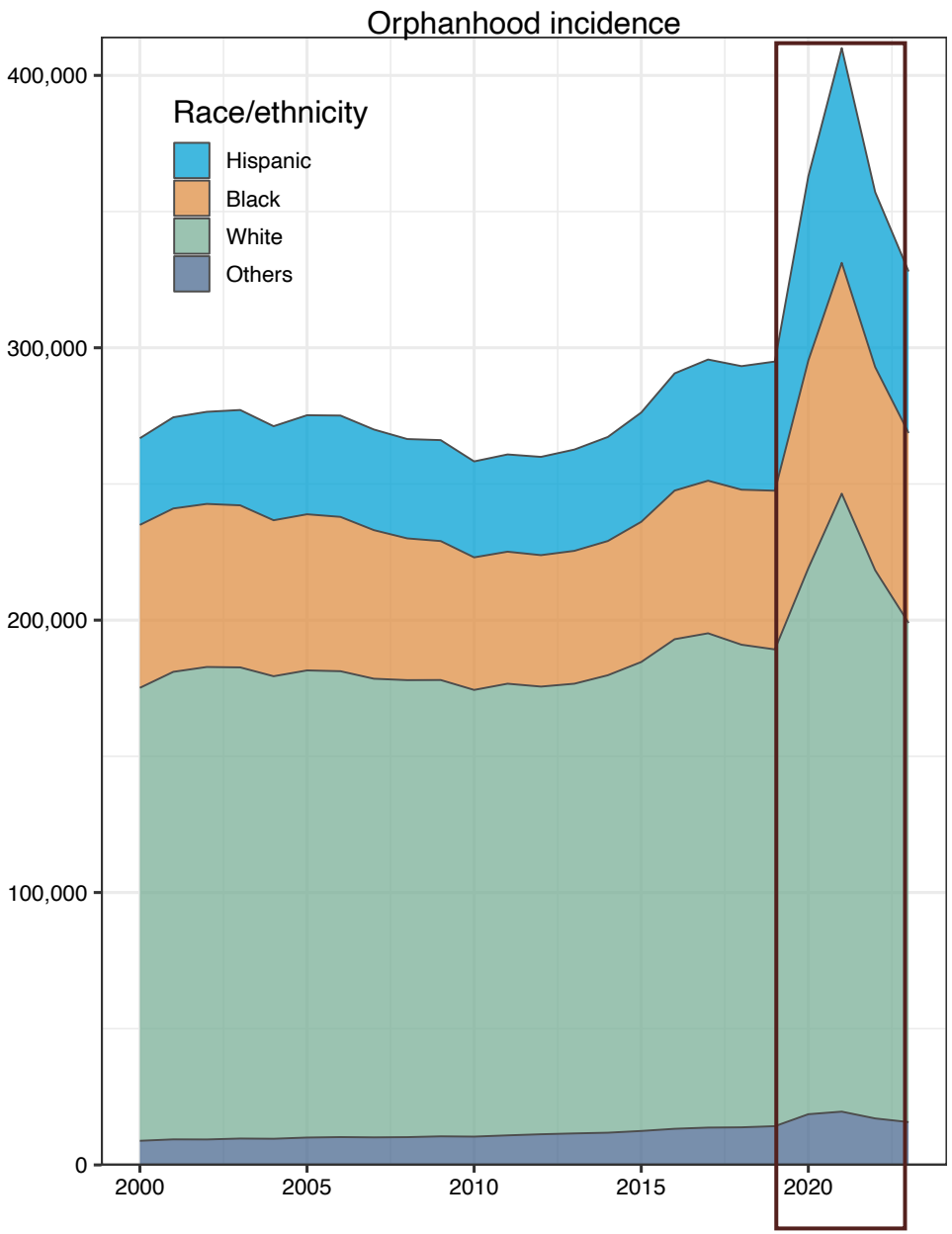
Annual number of orphanhood incidence by race/ethnicity, 2000-2023.

Heterogeneity in orphanhood across racial and ethnic groups.



Annual number of orphanhood incidence by race/ethnicity, 2000-2023.

Trends in parental loss had not recovered to pre-pandemic levels by 2023



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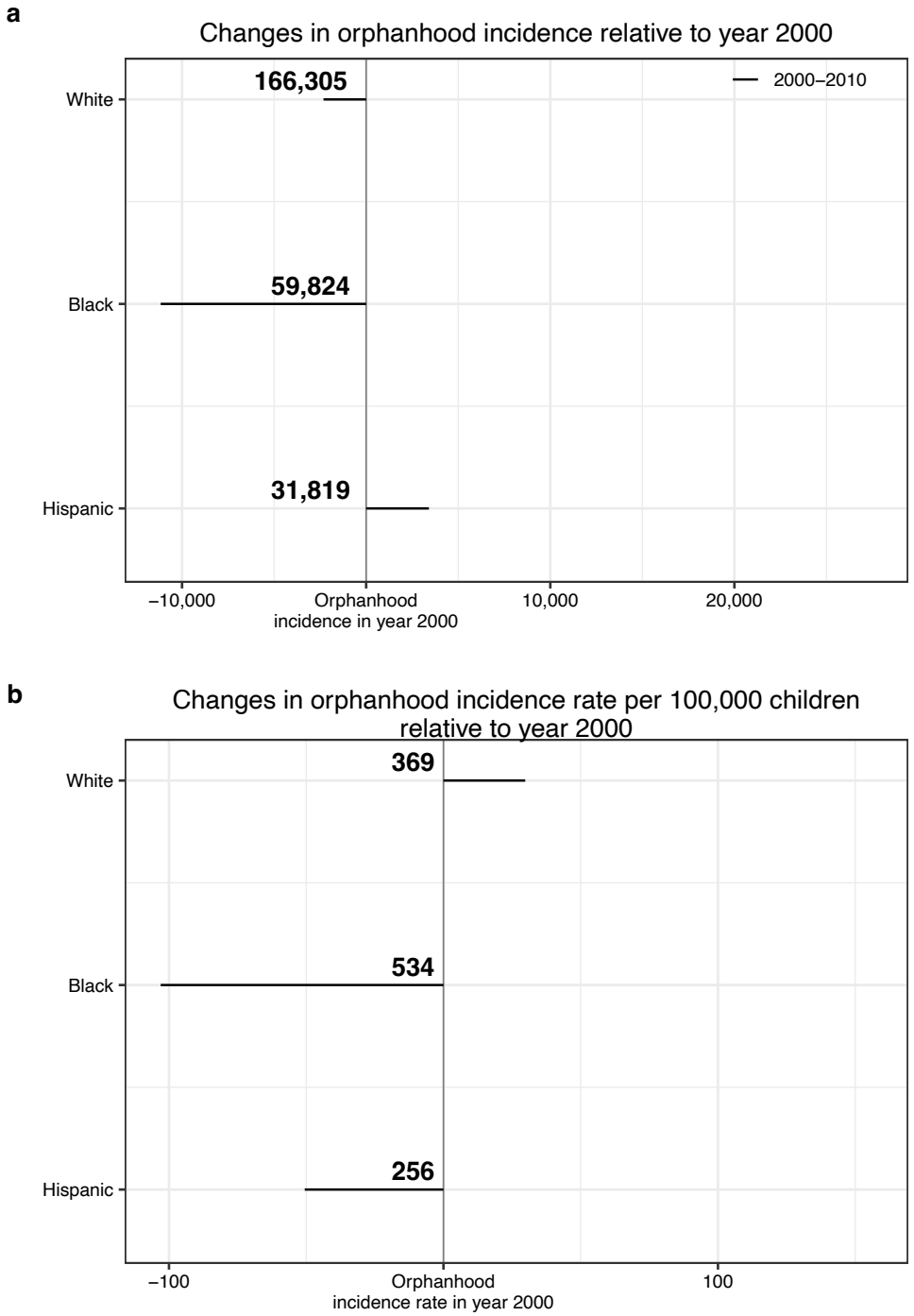
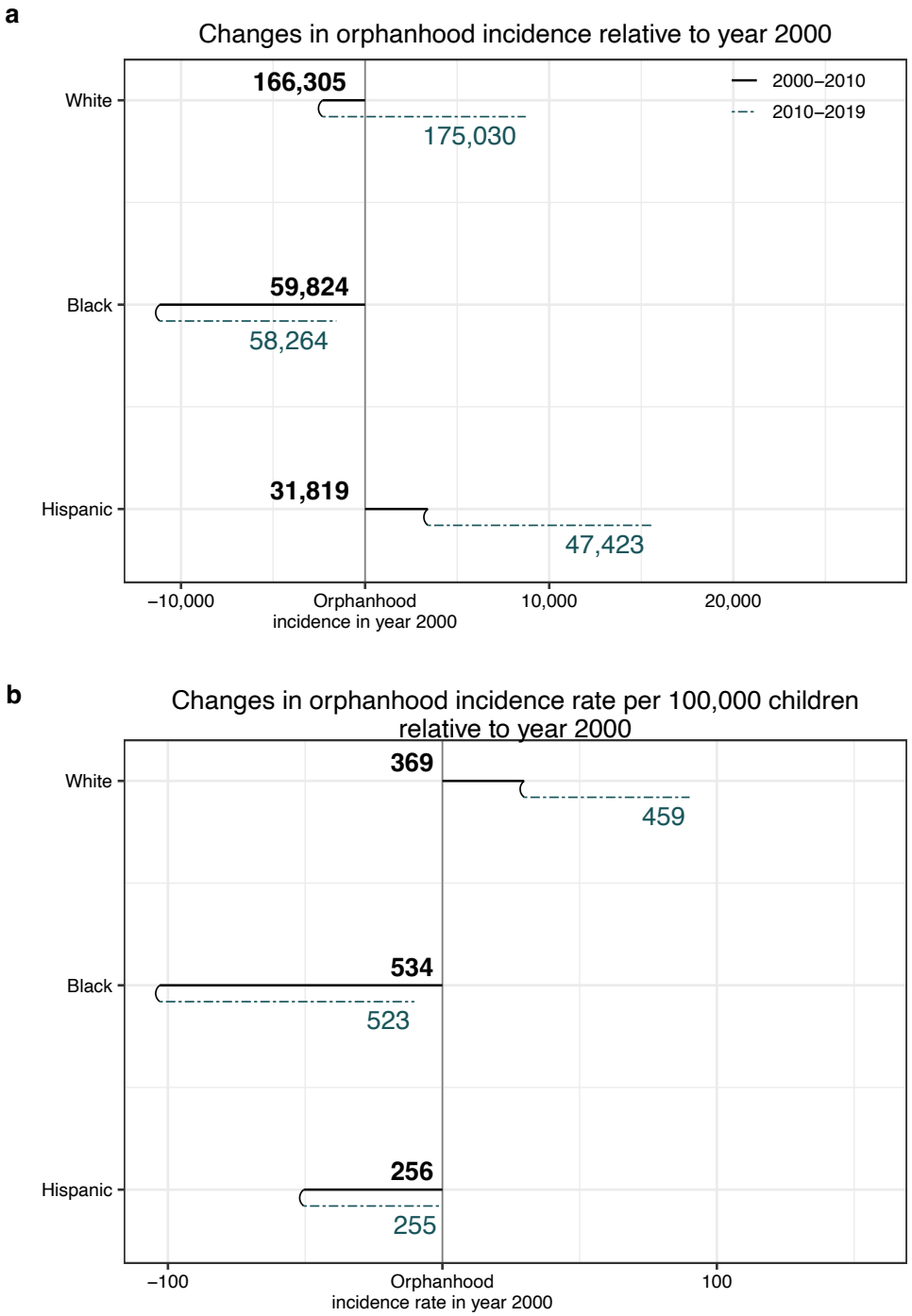
Orphanhood incidence changes in 2000—2010 by race and ethnicity.

2000-2010: no clear pattern

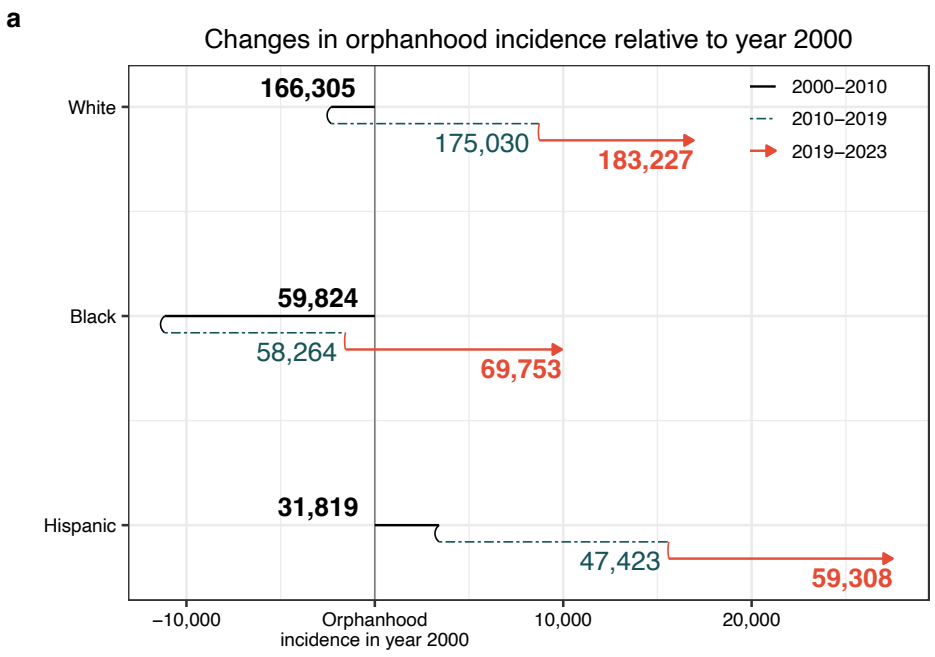


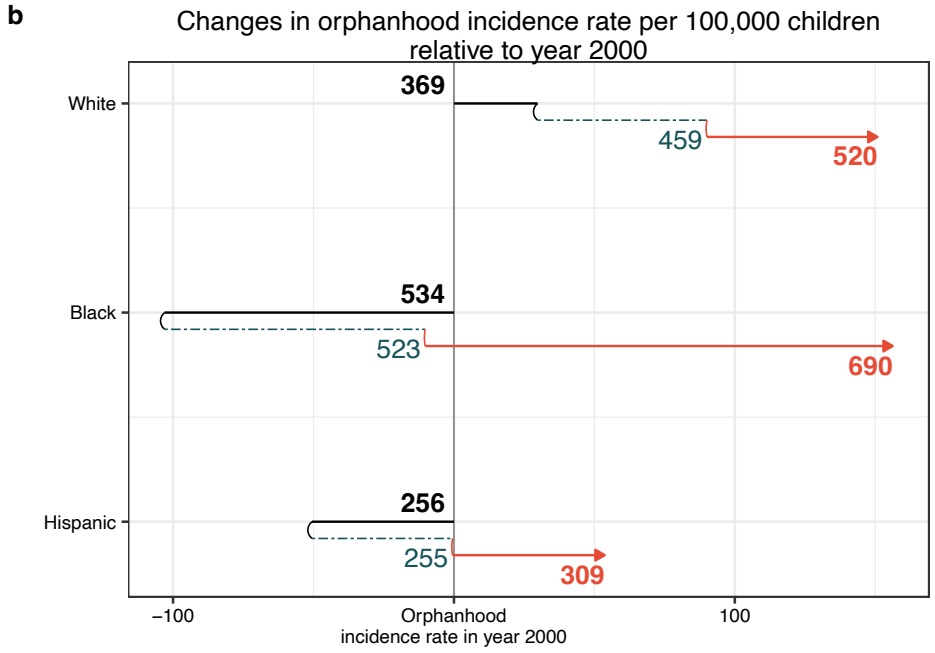
Orphanhood incidence changes in 2000—2010 and 2010—2019 by race and ethnicity.

2010-2019: increased orphanhood incidence counts and rates.



Orphanhood incidence changes in 2000—2010, 2010—2019, and 2019—2023 by race and ethnicity.





2019-2023: marked increase;

Substantial changes in incidence rates among Black.

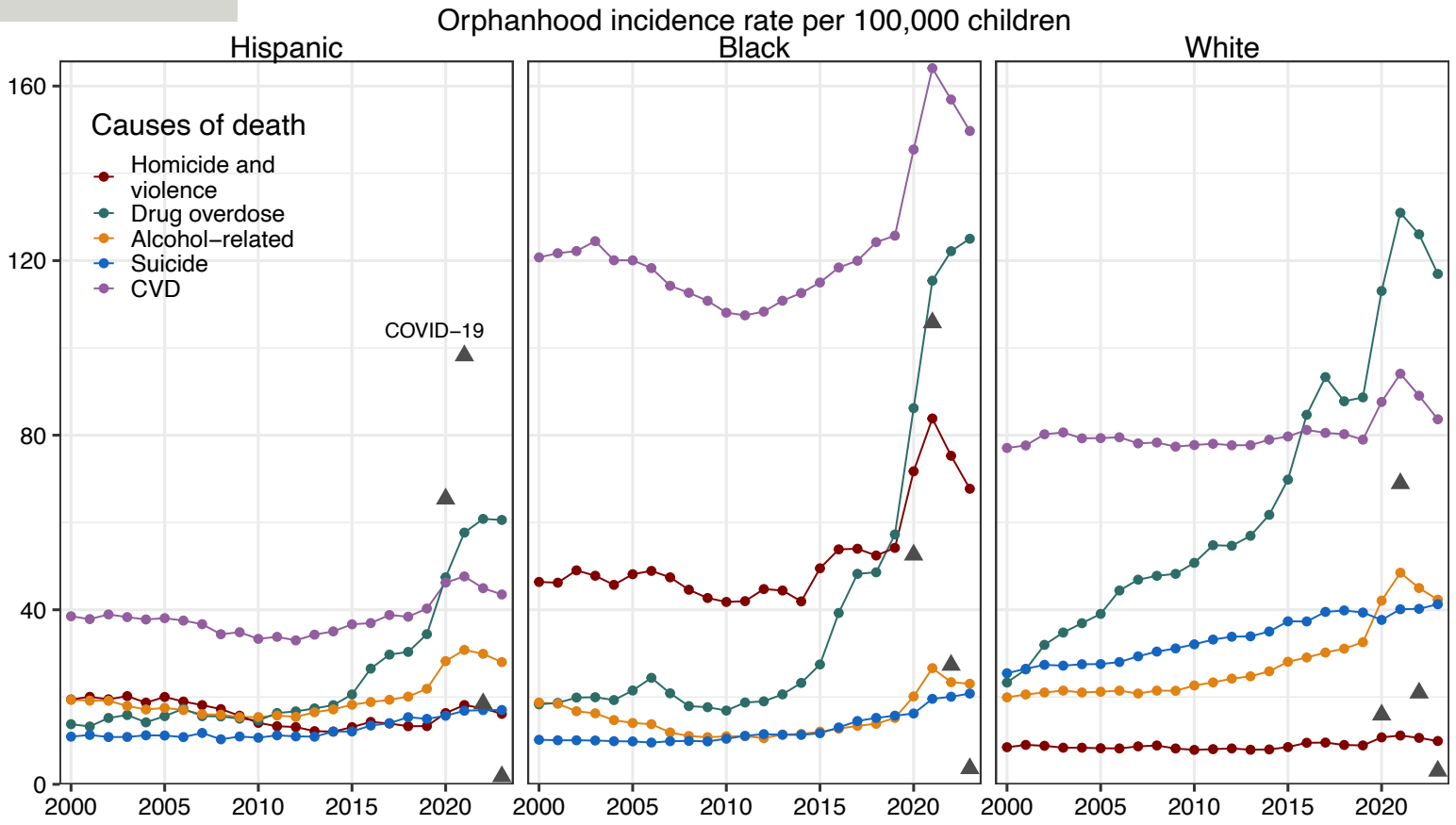
Annual orphanhood incidence rates by race/ethnicity, causes of death.

Primary drivers

CVD: all from 2015-2019, and Black 2000-2023

Drug overdose: Hispanic and White

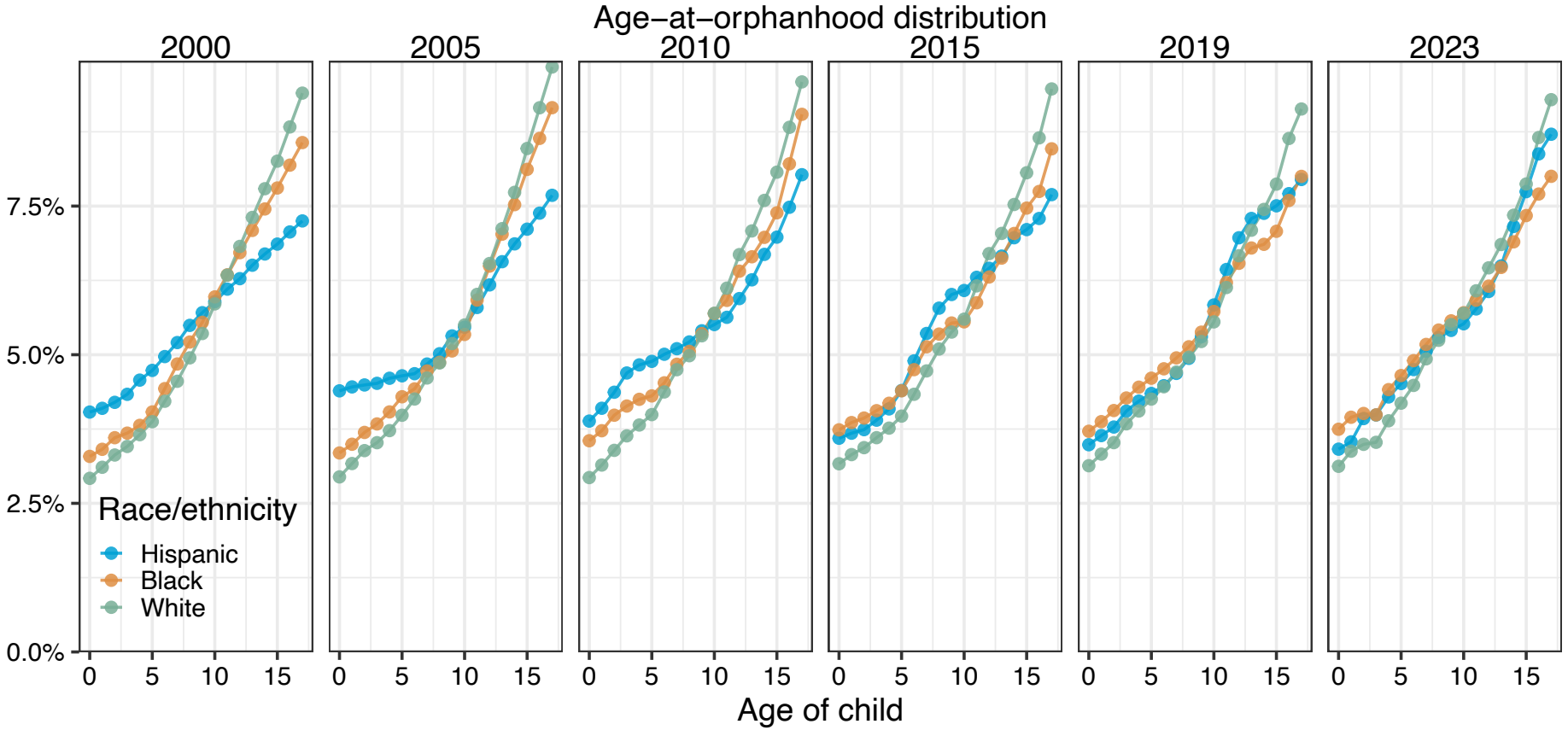
Homicide: Black (2nd and then 3rd)



The age-at-orphanhood distributions by race and ethnicity

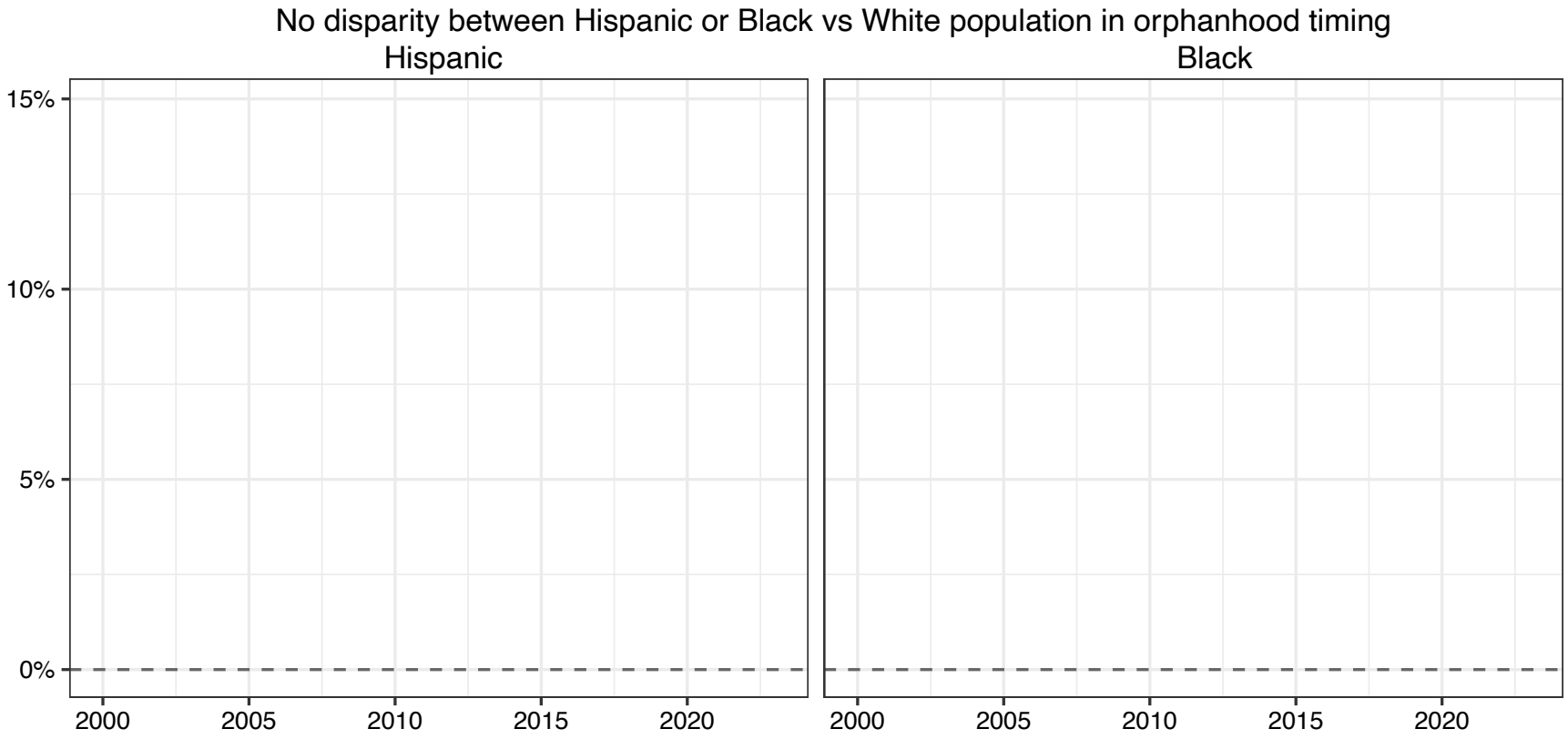
Age profiles among orphans

Whites as the reference group



Time trends in racial and ethnic disparity relative to Whites

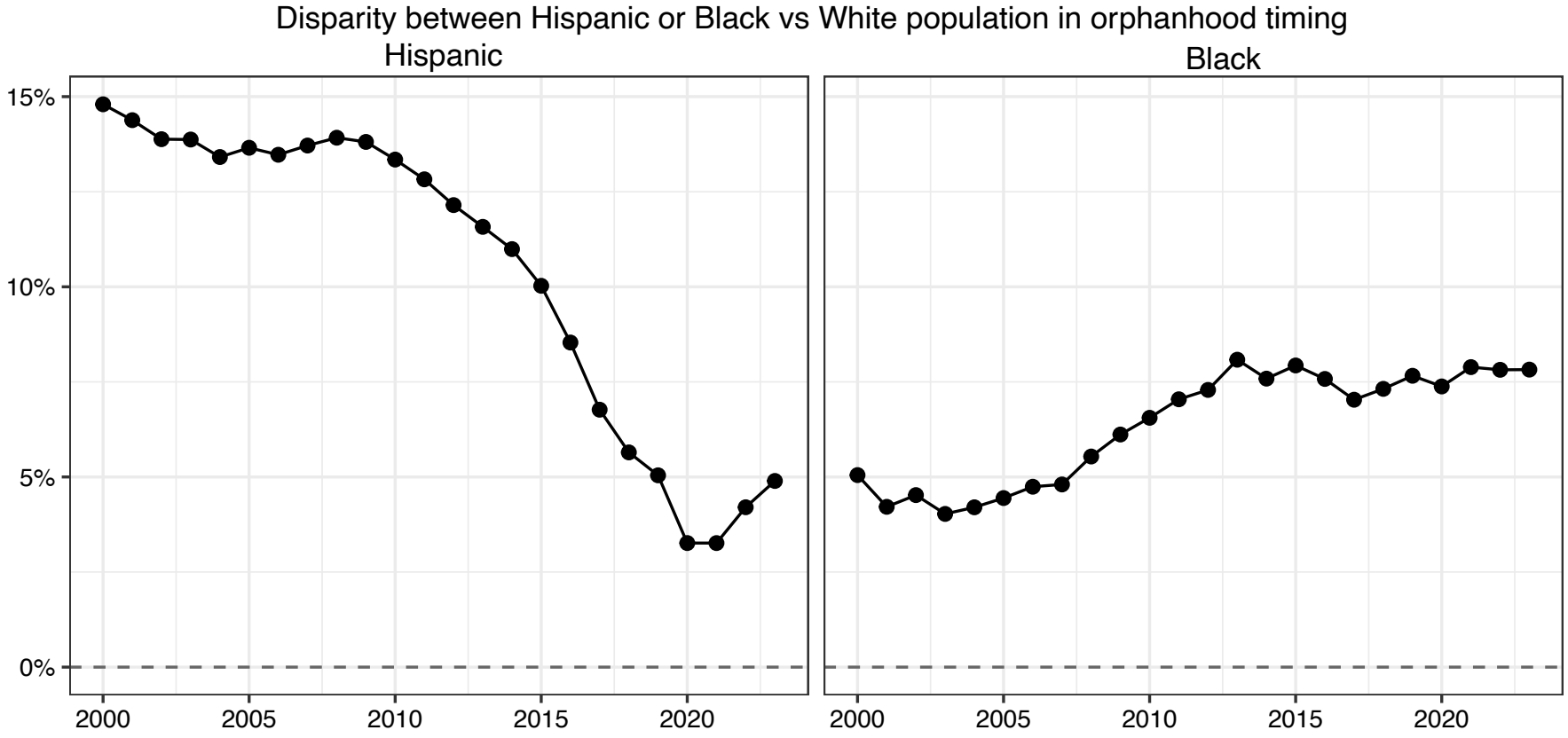
If there is no difference between
pairs of the age-at-orphanhood distributions



Time trends in racial and ethnic disparity relative to Whites

Quantified disparity between
pairs of the age-at-orphanhood distributions

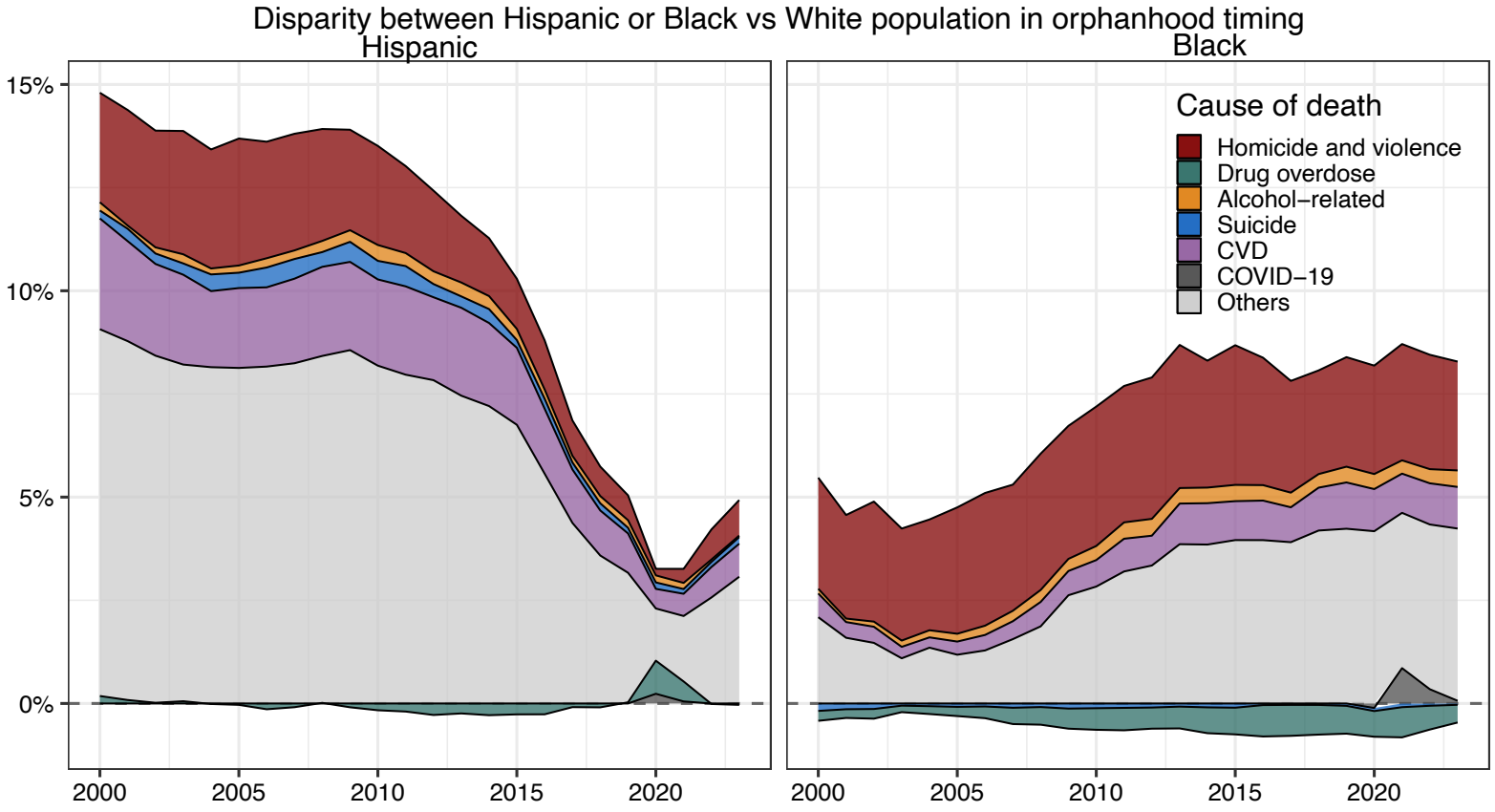
(Quantified S index)



Time trends in racial and ethnic disparity relative to Whites, with the cause-specific contributions using Shapley value decomposition.

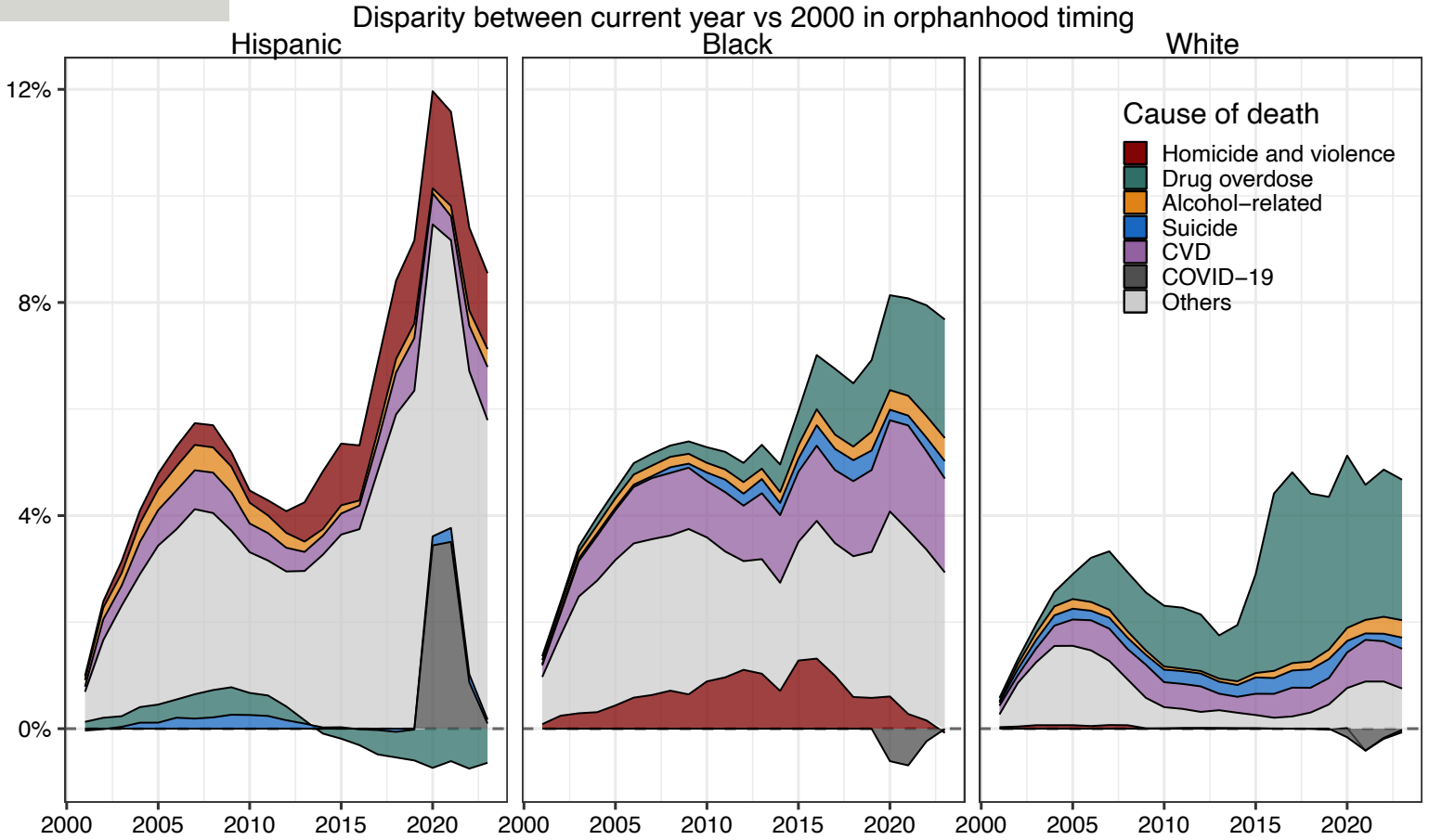
Different patterns across race/eth

Homicide and CVD as primary drivers
Drug overdose contributed negatively



Time trends in inequality measure of racial and ethnic subpopulations relative to 2000

Orphanhood timing changes significantly over time, with different patterns across race/ethnicity



Take away messages

- Orphanhood incidence:
  - **Increasing** trend over time before the pandemic;
  - **Not returned** to pre-pandemic levels by 2023;
  - **Racial and ethnic disparities**.

- Inequalities in orphanhood timing:
  - **Racial and ethnic disparities**;
  - Age-at-orphanhood distributions changed **over time**;
  - **Heterogeneity in the contribution of causes of death**.

Reference

- [1] Schluter, B. S., Alburez-Gutierrez, D., Bibbins-Domingo, K., Alexander, M. J. & Kiang, M. V. Youth experiencing parental death due to drug poisoning and firearm violence in the US, 1999–2020. JAMA 331, 1741–1747 (2024).
- [2] Verdery, A.M., Smith-Greenaway, E., Margolis, R. and Daw, J., 2020. Tracking the reach of COVID-19 kin loss with a bereavement multiplier applied to the United States. Proceedings of the National Academy of Sciences, 117(30), pp.17695-17701.
- [3] National Center for Health Statistics. Data Access - Vital Statistics Online (2023). Accessed Aug 3, 2023. https://www.cdc.gov/nchs/data_access/vitalstatsonline.htm.
- [4] Villaveces, A., Chen, Y. et al. Orphanhood and caregiver death among children in the United States by all-cause mortality, 2000–2021. Nature Medicine 31, 672–683 (2025) .
- [5] Shi, J., Aburto, J. M., Martikainen, P., Tarkiainen, L. & van Raalte, A. A distributional approach to measuring lifespan stratification. Population Studies 77, 15–33 (2023)
- [6] Shapley, L. S. et al. A value for n-person games (1953)

Advanced reading list

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Some more literature to compute number of orphanhood and grandparent caregivers

Check literatures:

Hillis, S. D., Unwin, H. J. T., Chen, Y., et al. "Global minimum estimates of children affected by COVID-19-associated orphanhood and deaths of caregivers: a modelling study." The Lancet 398.10298 (2021): 391-402.

Hillis, Susan D., et al. "COVID-19–associated orphanhood and caregiver death in the United States." Pediatrics 148.6 (2021): e2021053760.

Villaveces, A., Chen, Y., et al. "Orphanhood and caregiver death among children in the United States by all-cause mortality, 2000–2021." Nature Medicine 31.2 (2025): 672-683.

The work will be online soon!

Q & A

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https://www.lshtm.ac.uk/aboutus/people/chen.yu

Lecture notes and R labs:
https://github.com/YuCHENJT/PDP_2025-2026_lecture_materials